Electric Flux Density

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Gauss's Law

- The results of Faraday's experiments with the concentric spheres could be summed up as an experimental law by stating that the electric flux passing through any imaginary spherical surface lying between the two conducting spheres is equal to the charge enclosed within that imaginary surface.

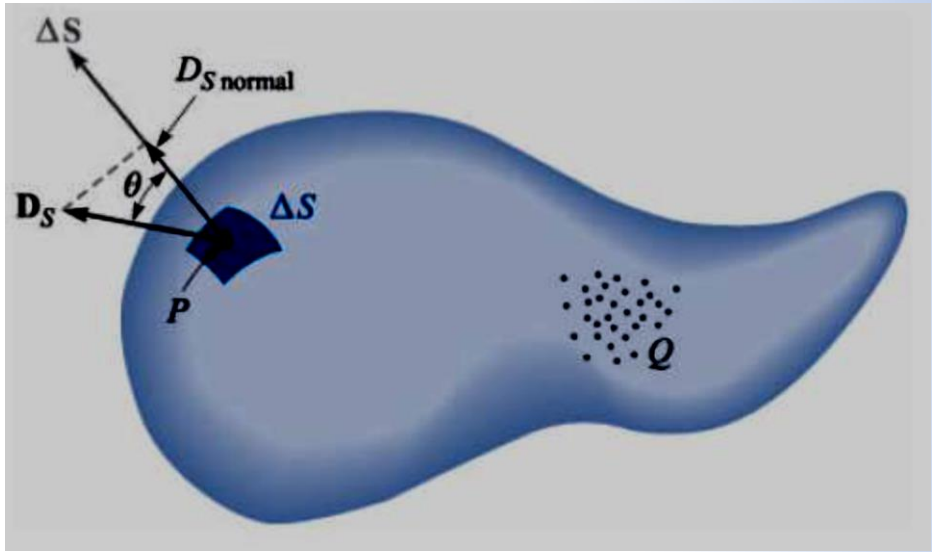
$$\psi = Q$$

- Faraday's experiment can be generalized to the following statement, which is known as Gauss's Law:

"The electric flux passing through <u>any</u> closed surface is equal to the total charge enclosed by that surface."

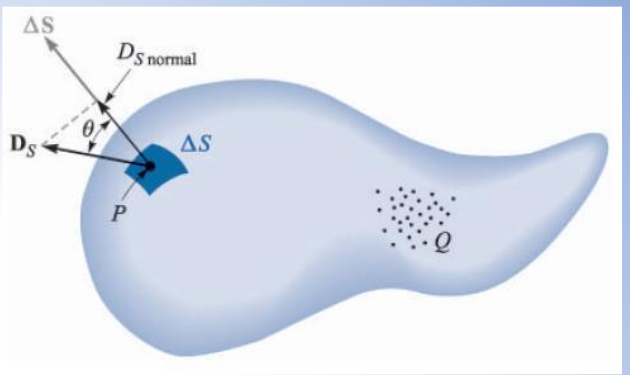
Gauss's Law

- Imagine a distribution of charge, shown as a cloud of point charges, surrounded by a closed surface of any shape.



- If the total charge is Q, the Q coulombs of electric flux will pass through the enclosing surface.
- At every point on the surface the electric-flux-density vector D will have some value D_S (subscript S means that D must be evaluated at the surface).

Gauss's Law



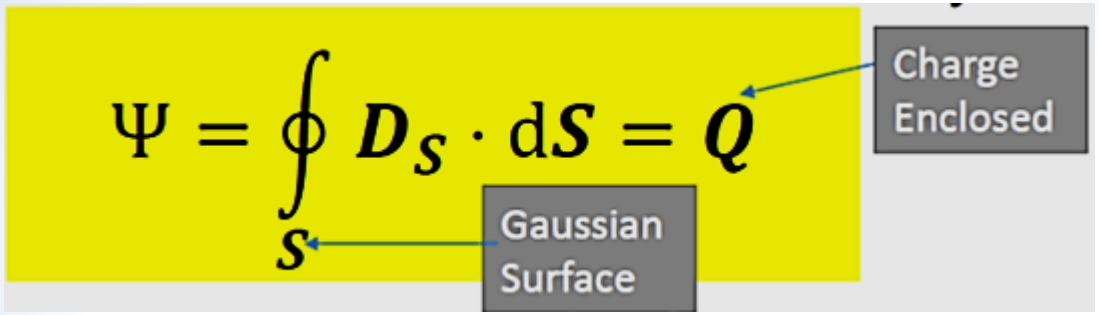
- ΔS defines an incremental element of area with magnitude of ΔS and the direction normal to the plane
- At any point P, where D_S makes an angle θ with ΔS, then the flux crossing ΔS is the product of the normal component of D_S and ΔS.

$$\Delta\Psi = flux\ crossing\ \Delta S = D_{S,norm}\Delta S$$
$$= D_s \cos\theta\ \Delta S$$
$$= \boldsymbol{D_s \cdot \Delta S}$$

$$\Psi = \int \boldsymbol{d}\Psi = \oint_{\boldsymbol{S}} \boldsymbol{D_S} \cdot \mathrm{d}\boldsymbol{S}$$

Gauss's Law

- We can formulate the Gauss's law mathematically as:



- The charge enclosed meant by the formula above might be several point charges, a line charge, a surface charge, or a volume charge distribution.

$$Q = \sum Q_n \qquad Q = \int \rho_L dL \qquad Q = \int_S \rho_S dS \qquad Q = \int_{vol} \rho_v dv$$

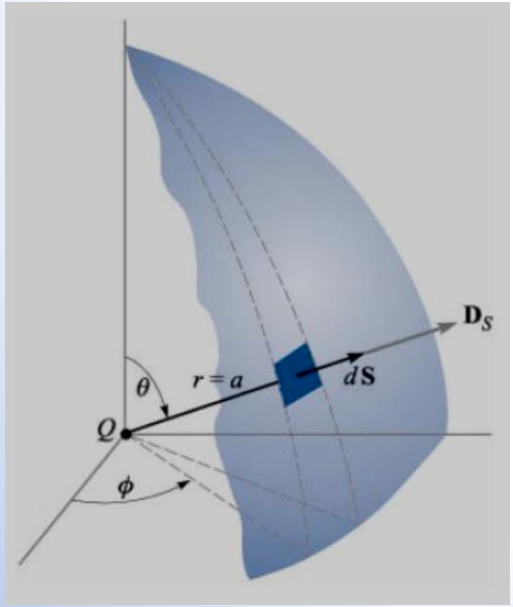
Gauss's Law

- We now take the last form, written in terms of the charge distribution, to represent the other forms:

$$\oint_S \boldsymbol{D}_S \cdot \mathbf{d}\boldsymbol{S} = \int_{vol} \rho_v \, \mathbf{d}v = Q$$

Gauss's Law

- Let a point charge Q be placed at the origin of a spherical coordinate system, and choose a closed surface as a sphere of radius a.
- The electric field intensity due to the point charge has been found to be:



$$\mathbf{E} = \frac{Q}{4\pi\varepsilon_0 r^2}\mathbf{a}_r$$

$$\mathbf{D} = \varepsilon_0\mathbf{E} \Rightarrow \mathbf{D} = \frac{Q}{4\pi r^2}\mathbf{a}_r$$

Gauss's Law

$$\mathbf{D}_S = \frac{Q}{4\pi a^2}\mathbf{a}_r$$

$$d\mathbf{S} = a^2 \sin\theta \, d\theta d\phi \, \mathbf{a}_r$$

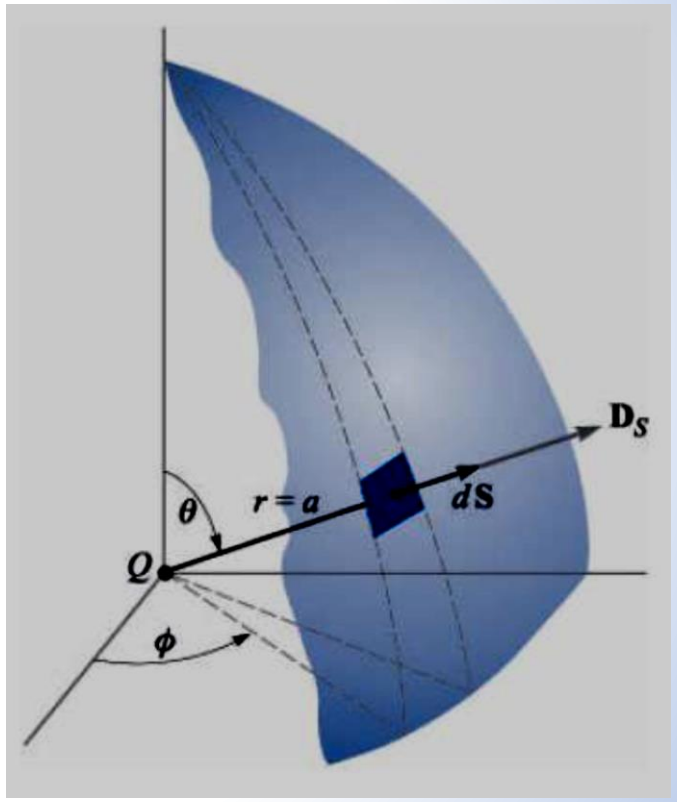
$$\mathbf{D}_S \cdot d\mathbf{S} = \frac{Q}{4\pi a^2} a^2 \sin\theta d\theta d\phi \, \mathbf{a}_r \cdot \mathbf{a}_r = \frac{Q}{4\pi}\sin\theta d\theta d\phi$$

$$\Psi = \oint_S \boldsymbol{D_S} \cdot \mathrm{d}\boldsymbol{S}$$

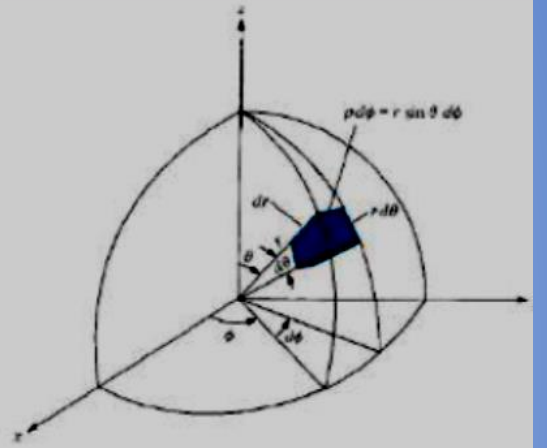
$$= \int_{\phi=0}^{2\pi}\int_{\theta=0}^{\pi} \frac{Q}{4\pi}\sin\theta d\theta d\phi \Big|_{r=a}$$

$$= -\frac{Q}{4\pi}\cos\theta\Big|_{\theta=0}^{\pi} \phi\Big|_{\phi=0}^{2\pi}$$

$$= Q$$



Spherical



$$d\boldsymbol{l} = dr\,\boldsymbol{a_r} + r\,d\theta\,\boldsymbol{a_\theta} + r\sin\theta\;d\phi\,\boldsymbol{a_\phi}$$

$$d\boldsymbol{S} = r^2\sin\theta\,d\theta\,d\phi\,\boldsymbol{a_r} = r\sin\theta\;dr\,d\phi\,\boldsymbol{a_\theta} = r\,dr\,d\theta\,\boldsymbol{a_\phi}$$

Application of Gauss's Law

- From the previous discussion of the uniform line charge, only the radial component of D is present:

$$\mathbf{D} = D_\rho \mathbf{a}_\rho$$

$$Q = \oint_S \mathbf{D}_S \cdot d\mathbf{S}$$

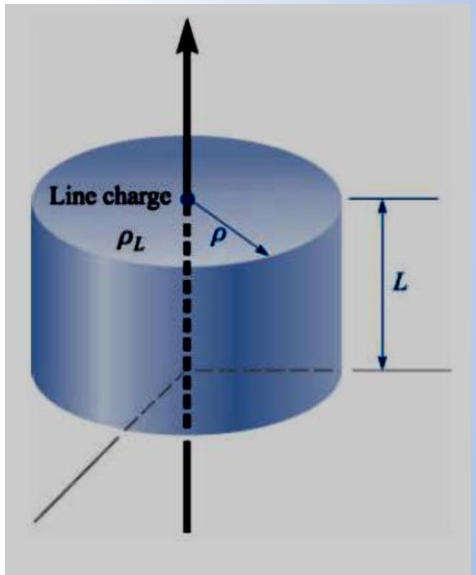
$$= D_\rho \int_{z=0}^{L} \int_{\phi=0}^{2\pi} \rho \, d\phi \, dz$$

$$= D_\rho 2\pi\rho L$$

$$Q = \rho_L L$$

$$D_\rho = \frac{\rho_L}{2\pi\rho}$$

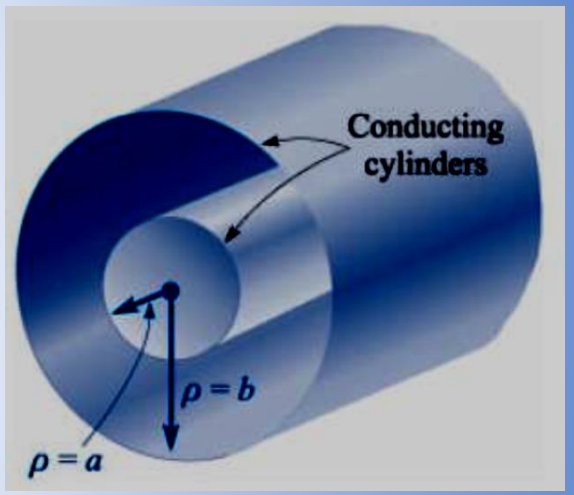
$$E_\rho = \frac{\rho_L}{2\pi\varepsilon_0\rho}$$



Application of Gauss's Law

- The problem of a coaxial cable is almost identical with that of the line charge.
- Suppose that we have two coaxial cylindrical conductors, the inner of radius a and the outer of radius b, both with infinite length.
- We shall assume a charge distribution of ρ_S on the outer surface of the inner conductor.
- Choosing a circular cylinder of length L and radius ρ, $a < \rho < b$, as the gaussian surface, we find:

$$Q = D_S 2\pi\rho L$$



- The total charge on a length L of the inner conductor is:

$$Q = \int_{z=0}^{L}\int_{\phi=0}^{2\pi} \rho_S a\, d\phi\, dz = 2\pi a L \rho_S \Rightarrow \boxed{D_S = \frac{a\rho_S}{\rho}}$$

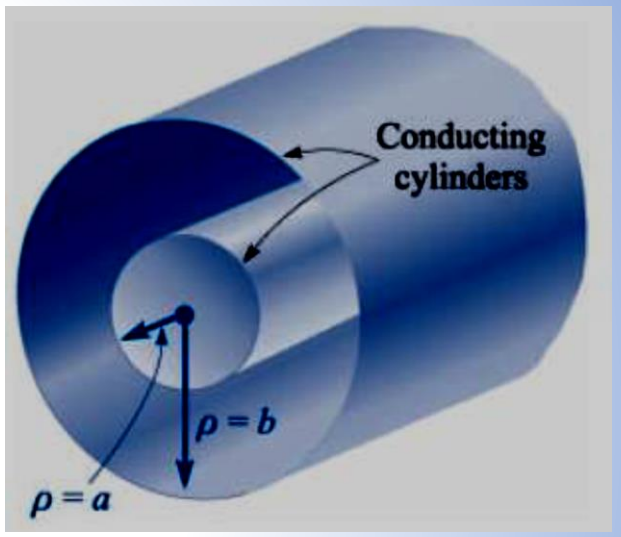
Application of Gauss's Law

$$D_s = \frac{a\rho_s}{\rho}$$

- For one meter length, the inner conductor has $2\pi a\rho_S$ coulombs, hence $\rho_L = 2\pi a\rho_S$,

$$\mathbf{D} = \frac{\rho_L}{2\pi\rho}\mathbf{a}_\rho$$

a < ρ < b,



- Every line of electric flux starting from the inner cylinder must terminate on the inner surface of the outer cylinder:

$$Q_{\text{outer cyl}} = -2\pi aL\rho_{S,\text{inner cyl}}$$

$$2\pi bL\rho_{S,\text{outer cyl}} = -2\pi aL\rho_{S,\text{inner cyl}}$$

$$\rho_{S,\text{outer cyl}} = -\frac{a}{b}\rho_{S,\text{inner cyl}}$$

- If we use a cylinder of radius $\rho > b$, then the total charge enclosed will be zero.
  - ► There is no external field, $D_s = 0$

An identical result would be obtained for $\rho < a$.

Application of Gauss's Law

■ **Example**

A 50-cm length of coaxial cable has an inner radius of 1 mm and an outer radius of 4 mm. The space between conductors is assumed to be filled with air. The total charge on the inner conductor is 30 nC. Find the charge density on each conductor and the expressions for E and D fields.

$$Q_{\text{inner cyl}} = 2\pi a L \rho_{S,\text{inner cyl}}$$

$$\Rightarrow \rho_{S,\text{inner cyl}} = \frac{Q_{\text{inner cyl}}}{2\pi a L} = \frac{30\times10^{-9}}{2\pi(10^{-3})(0.5)} = \underline{\underline{9.55\,\mu\text{C/m}^2}}$$

$$Q_{\text{outer cyl}} = 2\pi b L \rho_{S,\text{outer cyl}} = -Q_{\text{inner cyl}}$$

$$\Rightarrow \rho_{S,\text{outer cyl}} = \frac{-Q_{\text{inner cyl}}}{2\pi b L} = \frac{-30\times10^{-9}}{2\pi(4\times10^{-3})(0.5)} = \underline{\underline{-2.39\,\mu\text{C/m}^2}}$$

$$D_\rho = a\frac{\rho_{S,\text{inner cyl}}}{\rho} = 10^{-3}\frac{(9.55\times10^{-6})}{\rho} = \underline{\underline{\frac{9.55}{\rho}\text{nC/m}^2}}$$

$$E_\rho = \frac{D_\rho}{\varepsilon_0} = \frac{9.55\times10^{-9}}{8.854\times10^{-12}\rho} = \underline{\underline{\frac{1079}{\rho}\text{V/m}}}$$

Both of these expressions apply to the region where $1 < \rho < 4$ mm. For $\rho < 1$ mm or $\rho > 4$ mm, **E** and **D** are zero.


Thank
you